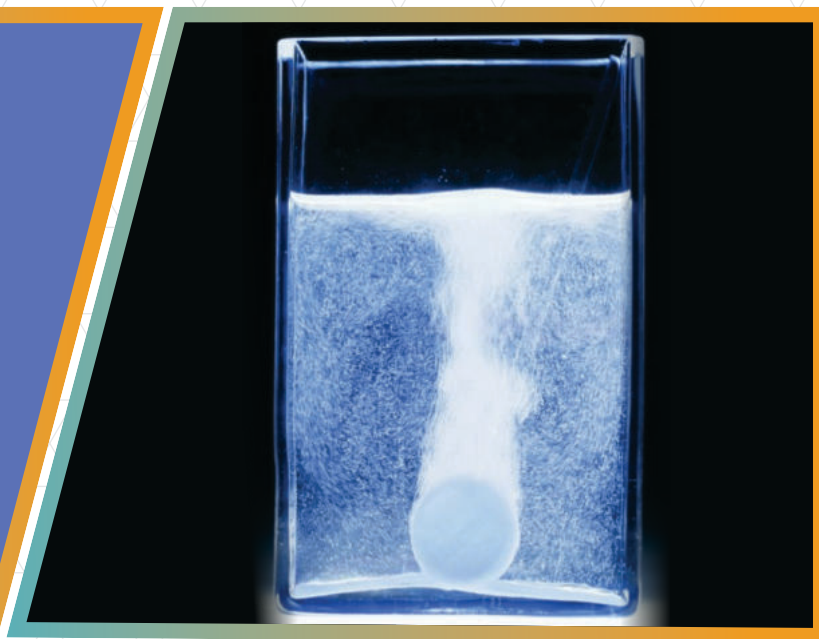




UNIT 06

SOLUTIONS



A solution is a homogeneous mixture of two or more substances, one with no boundaries separating its components; in other words, a solution exists as one phase.

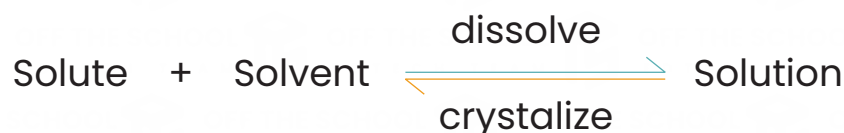
COMPONENTS OF A SOLUTION:

i. SOLVENT: it is the component of a solution always present in a greater amount.

ii. SOLUTE: it is the component of a solution always present in a smaller amount.

Explanation:

A solute dissolve in a solvent to form a solution.



AQUEOUS SOLUTION:

That type of solution in which solvent is always water.

Examples: Tang, sugar, salt and many other solutes which can be dissolve in water to obtain the aqueous solutions.

TYPES OF SOLUTION:

(i) Saturated Solution: A solution which contains maximum amount of a solute that can be dissolved in a solvent at a given temperature is called saturated solution at that particular temperature. This solution can no longer dissolve any more solute under the given conditions.

(ii) Unsaturated Solution: The solution containing lesser amount of the solute than the saturated solution at a given temperature is called an unsaturated solution. In the unsaturated solution, the solvent has the capacity of dissolving more amount of the solute at that particular temperature.

(iii) Supersaturated Solution: A solution that can dissolve more solute than it contained in the saturated solution after heating is supersaturated solution.



(a)



(b)



(c)



(d)



(e)

▲ (a) Many solutions consist of a solid dissolved in water. (b) When all the solid dissolves, the solution is unsaturated. (c) If more solid is added than will dissolve, the solution is saturated. (d) A saturated solution is, by definition, in contact with undissolved solid. (e) Some saturated solutions can be made into supersaturated solutions by heating to dissolve more solid, and cooling carefully to prevent crystallization.

DILUTION OF A SOLUTION:

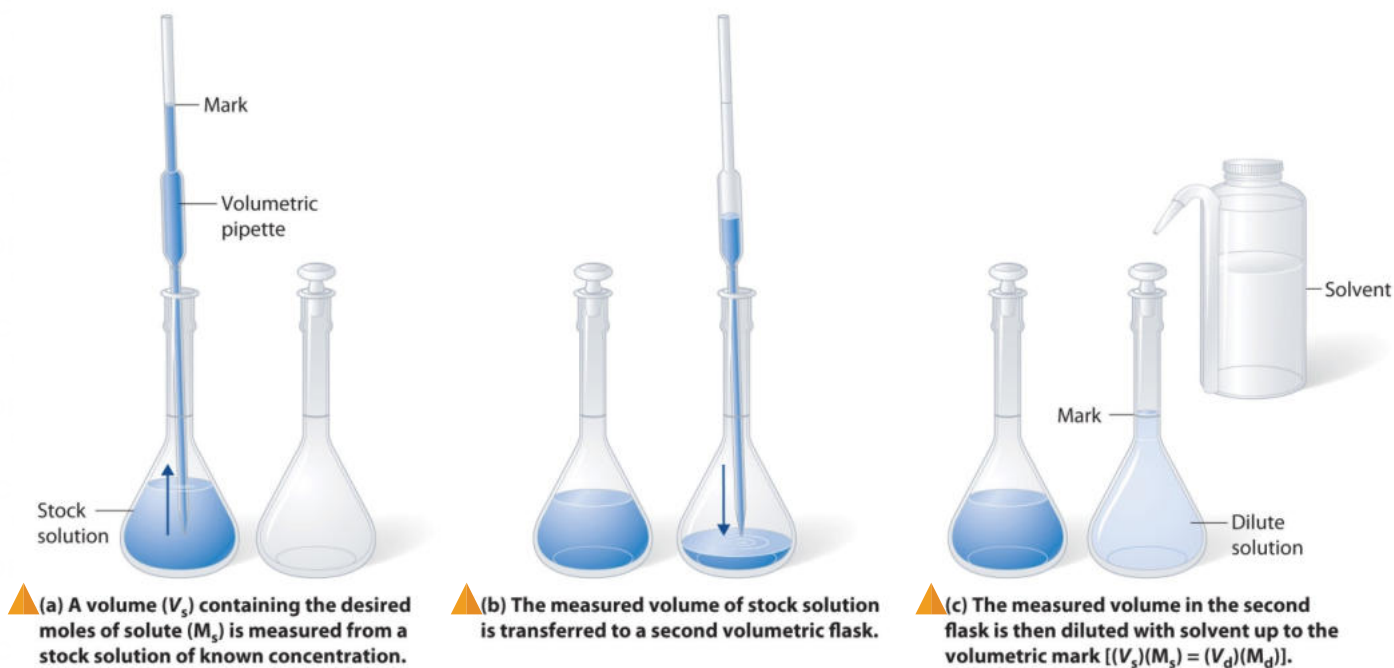
Dilution is the opposite to concentration (means to reduce the concentration of any substance is known as dilution).

DILUTE SOLUTION: the solution in which solute is present relatively in small amount.

CONCENTRATION SOLUTION: the solution in which solute is present relatively in large amount.

Concentrated solution = Diluted solution

$$\therefore M_1V_1 = M_2V_2 \quad \text{or} \quad M_cV_c = M_dV_d$$

**CLASSIFICATION OF A SOLUTION**

Types of Solutions			
Solute	Solvent	State of Resulting Solution	Example
Gas	Gas	Gas	Air
Gas	Liquid	Liquid	Carbonated water
Gas	Solid	Solid	H ₂ gas in palladium
Liquid	Liquid	Liquid	Ethanol in water
Liquid	Solid	Solid	Mercury in silver
Solid	Liquid	Liquid	Saltwater
Solid	Solid	Solid	Brass (Cu/Zn)

MOLARITY:

Molarity, M , is defined as the number of moles of solute divided by the number of liters of solution.

$$\therefore \text{Molarity} = M = \frac{\text{moles}}{\text{volume in litres}} = \frac{n}{V}$$

MOLE FRACTION:

Mole fraction, χ , which is defined as the number of moles of solute divided by the total number of moles of solution, is also an expression of concentration.

$$\therefore \text{Mole fraction of component A} = \chi_A = \frac{\text{moles of A}}{\text{sum of moles of all components}} = \frac{n_A}{n_T}$$

MOLALITY:

Molality (m) is the number of moles of solute dissolved in 1 kg (1000 g) of solvent:

$$\therefore \text{Molality} = m = \frac{\text{moles of solute}}{\text{mass of solvent (in kg)}}$$

SOLUBILITY:

Solubility is defined as the number of grams of the solute dissolved in 100 g of a solvent to prepare a saturated solution at a particular temperature.

Following are the factors which affect the solubility of solutes:



▲ Piece of sugar is lifted in water and dissolve gradually.

1. The general principle of solubility is, like dissolves like.

The principle "like dissolves like" means that substances with similar properties can dissolve in each other. For example:

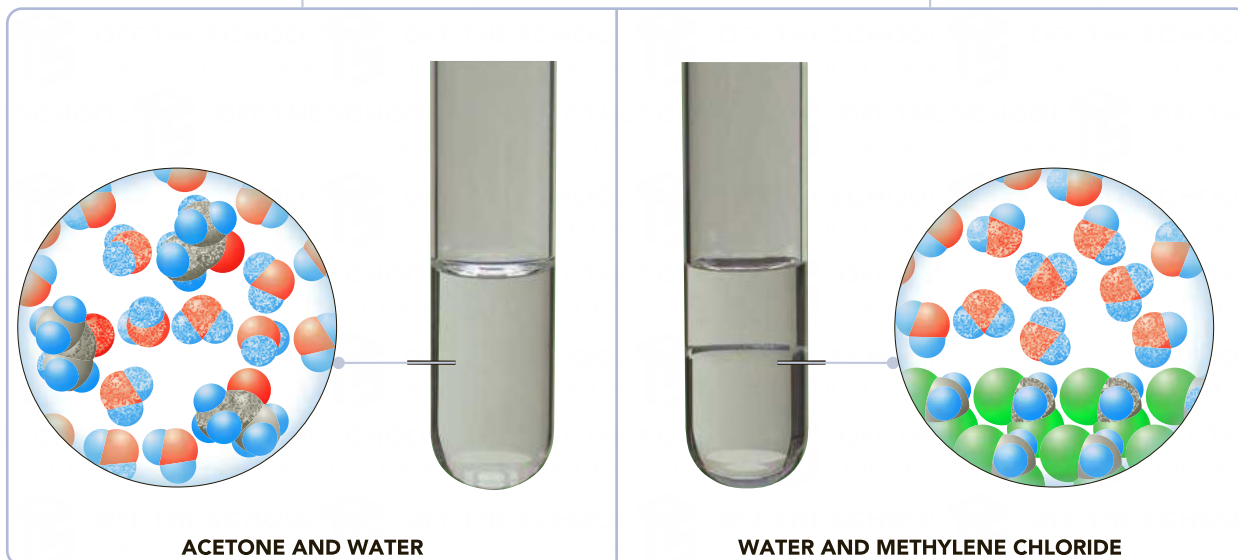
- ✓ **Polar substances** (like water) can dissolve other polar substances (like salt).
- ✓ **Nonpolar substances** (like oil) can dissolve other nonpolar substances (like wax).

In simple terms, substances that are alike in their chemical nature will mix or dissolve well together.

▼ Immiscible and miscible liquids

Acetone and water are miscible liquids; that is, the two substances dissolve in each other in all proportions.

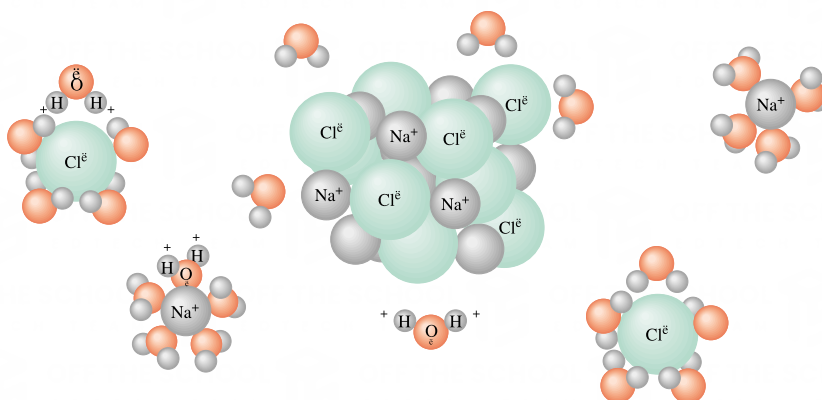
Water and methylene chloride are immiscible and form two layers.



2. Solute-solvent interaction:

To dissolve one substance (solute) in another substance (solvent) following three events must occur:

- (i) Solute particles separation
- (ii) Solvent particles separation to provide space for solute particles.
- (iii) Solute and solvent particles attraction.



▲ **Process of dissolving NaCl (salt) in water:** The image shows how water molecules separate the Na^+ and Cl^- ions from the salt crystal and attract them to complete the dissolution process.

3. Temperature:

The effect of temperature is same in solid and liquid but different in gases.

- i) **For solids and liquids:** solubility is directly proportional to the temperature in solids and liquids.
- ii) **For gases:** gas solubility typically decreasing as temperature increases.

SUSPENSION:

If the particles in a solvent are so large that they settle out unless the mixture is constantly stirred or agitated, the mixture is called a suspension.

Explanation:

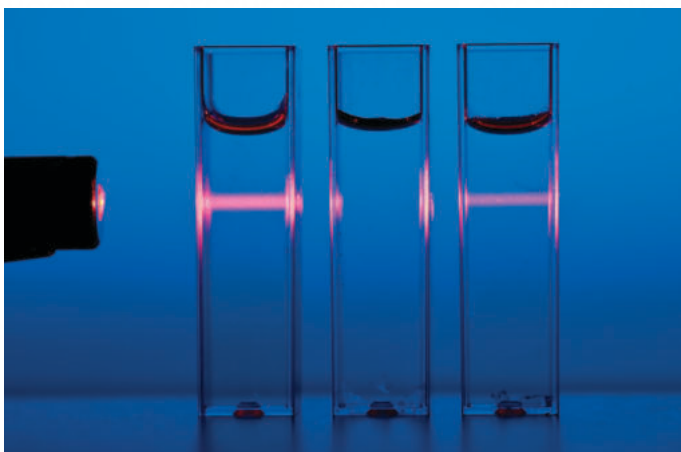
Think of a jar with oil and water. If left undisturbed, the oil particles float on top of the water, but they don't mix. Similarly, in a suspension, solid particles (like soil or sand) are spread out in a liquid but will eventually settle at the bottom if the mixture is left still. These particles are large enough to be seen without a microscope, and they can be separated using a filter. Just like oil and water don't mix, particles in a suspension can be separated because they don't dissolve into the liquid.

**COLLOIDS:**

Particles that are intermediate in size between those in solutions and suspensions form mixtures known as colloidal dispersions, or simply colloids.

TYNDALLEFFECT:

When a beam of light rays is passed through a colloid, particles of the dispersed phase scatter the light rays and the path of the light becomes visible. The scattering of light by the colloid particles is known as Tyndall effect.



▲ The Tyndall effect. Light is scattered by colloidal particles (left and right) but not by dissolved particles (center).



▲ A familiar example of the Tyndall effect: headlights illuminating fog.

ProPertieS of SolutionS, ColloidS, and SuSPenSionS

Solutions	Colloids	Suspensions
Homogeneous	Heterogeneous	Heterogeneous
Particle size: 0.01–1 nm; can be atoms, ions, molecules	Particle size: 1–1000 nm, dispersed; can be aggregates or large molecules	Particle size: over 1000 nm, suspended; can be large particles or aggregates
Do not separate on standing	Do not separate on standing	Particles settle out
Cannot be separated by filtration	Cannot be separated by filtration	Can be separated by filtration
Do not scatter light	Scatter light (Tyndall effect)	May scatter light, but are not transparent

NUMERICAL PROBLEMS

MOLARITY:

1. What is the molarity of a solution that contains 0.2 moles of solute in 500 ml of solvent?
2. How many moles of solute are present in a 2 M solution that contains 100 ml of solvent?
3. If you have 50 ml of a 3 M solution, how many moles of solute are present in it?
4. What is the volume of solvent required to prepare a 0.5 M solution that contains 2 moles of solute?
5. If you have 0.3 moles of solute and need to prepare a 0.1 M solution, what volume of solvent should you use?
6. Calculate the number of moles of solute in a 50 mL solution containing 0.1 M of sodium chloride.
7. What is the molarity of a solution containing 0.5 moles of solute in 250 mL of solution?
8. How many milliliters of a 2 M solution of hydrochloric acid are needed to prepare 0.5 L of a 0.5 M solution?
9. A solution contains 10 g of solute in 500 mL of solution. What is its molarity?
10. What is the mass of solute in 500 mL of a 1 M solution of glucose?

DILUTION OF SOLUTION:

1. A stock solution of hydrochloric acid (HCl) has a concentration of 10 M. How much of this stock solution is required to prepare 500 mL of a 2 M HCl solution?
2. If you need to prepare 200 mL of a 0.5 M NaOH solution from a 2 M NaOH stock solution, how much of the stock solution would you need?
3. How much water should be added to 100 mL of a 6 M sulfuric acid (H_2SO_4) solution to dilute it to 1 M?
4. You have a 4 M potassium chloride (KCl) solution. How much of the solution is required to prepare 1.5 L of a 0.8 M KCl solution?
5. To prepare 250 mL of a 0.2 M ethanol solution from a stock solution of 1 M ethanol, how much of the stock solution should be used?

MISCELLANEOUS:

1. A solution contains 20 g of sodium chloride (NaCl) dissolved in 200 mL of water. What is the percentage concentration (m/v) of the solution?
2. A solution of ethanol contains 30 g of ethanol dissolved in 150 mL of water. What is the percentage concentration (m/v) of the ethanol solution?
3. How much sodium hydroxide (NaOH) is required to prepare 250 mL of a 0.2M NaOH solution?
4. What volume of concentrated HCl (37%) is required to prepare 500 mL of a 10% (v/v) solution?
5. A 400 cm^3 solution contains 5.6 g of magnesium chloride (MgCl_2). Find the concentration in g/dm^3 .
6. How much of 0.3M hydrochloric acid is required to neutralize 50 mL of 0.5M sodium hydroxide solution?